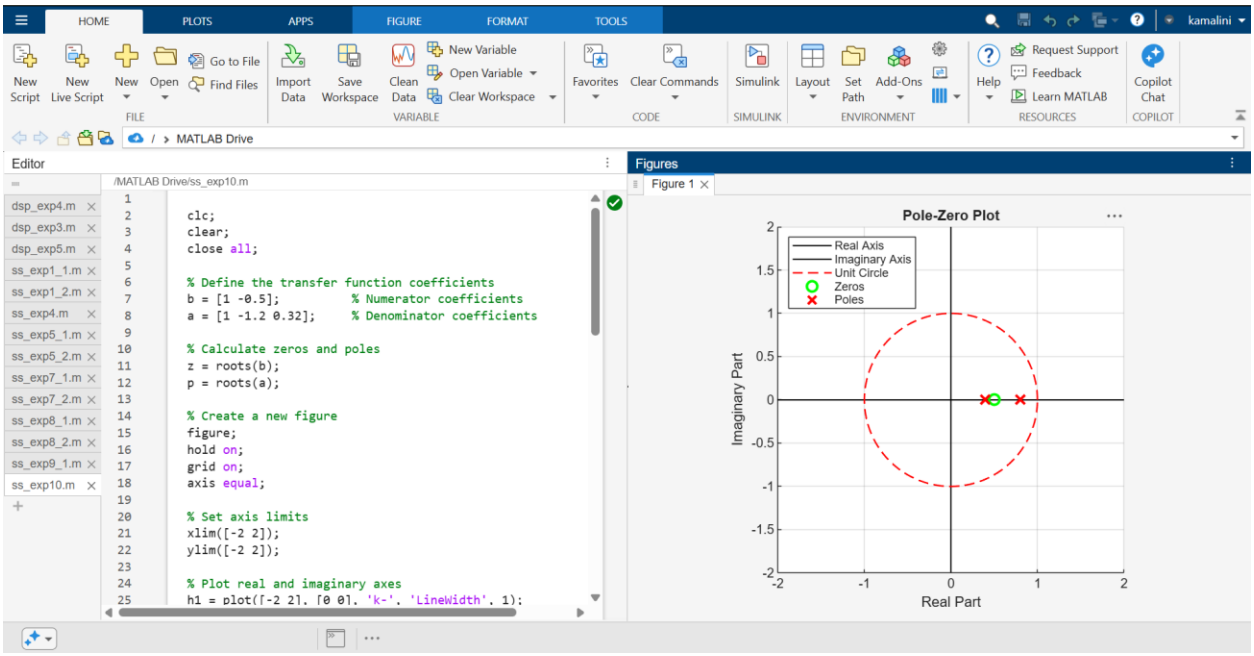
EXPERIMENT 10

```
clc;
clear;
close all;
% Define the transfer function coefficients
b = [1 -0.5]; % Numerator coefficients
a = [1 -1.2 0.32]; % Denominator coefficients
% Calculate zeros and poles
z = roots(b);
p = roots(a);
% Create a new figure
figure;
hold on;
grid on;
axis equal;
% Set axis limits
xlim([-2 2]);
ylim([-2 2]);
% Plot real and imaginary axes
h1 = plot([-2 2], [0 0], 'k-', 'LineWidth', 1);
h2 = plot([0 0], [-2 2], 'k-', 'LineWidth', 1);
% Plot the unit circle
theta = linspace(0, 2*pi, 200);
h3 = plot(cos(theta), sin(theta), 'r--', 'LineWidth', 1.2);
% Plot zeros as green circles
h4 = plot(real(z), imag(z), 'go', ...
'MarkerSize', 8, 'LineWidth', 2);
% Plot poles as red crosses
h5 = plot(real(p), imag(p), 'rx', ...
'MarkerSize', 10, 'LineWidth', 2);
% Customize the plot
xlabel('Real Part');
ylabel('Imaginary Part');
title('Pole-Zero Plot');
legend([h1 h2 h3 h4 h5], ...
{'Real Axis', 'Imaginary Axis', 'Unit Circle', ...
'Zeros', 'Poles'}, ...
'Location', 'northwest');
hold off;
```


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clc;
clear;
close all;
% Define the transfer function coefficients
b = [1 -0.5]; % Numerator coefficients
a = [1 -1.2 0.32]; % Denominator coefficients
% Calculate zeros and poles
z = roots(b);
p = roots(a);
% Create a new figure
figure;
hold on;
grid on;
axis equal;
% Set axis limits
xlim([-2 2]);
ylim([-2 2]);
% Plot real and imaginary axes
h1 = plot([-2 2], [0 0], 'k-', 'LineWidth', 1);
Figures
Figure 1
Pole-Zero Plot
Real Axis
Imaginary Axis
Unit Circle
Zeros
Poles
Imaginary Part
Real Part
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2
-1
0
1
2